Proceedings of the Third International Conference on Smart Electronics and Communication (ICOSEC 2022)
IEEE Xplore Part Number: CFP22V90-ART; ISBN: 978-1-6654-9764-0

2022 3rd International Conference on Smart Electronics and Communication (ICOSEC) | 978-1-6654-9764-0/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICOSEC54921.2022.9951937

Performance Evaluation of Deep Learning based Mandarin Fruit Sorting System with Industrial Camera

Muralidharan Duraisamy
Dept. of Electronics & Communication Engineering
Jeppiaar Engineering College
Chennai, Tamil Nadu, India
muralidharanduraisamy2001@gmail.com

Rajavendhan Govindaraj
Dept. of Electronics & Communication Engineering
Jeppiaar Engineering College
Chennai, Tamil Nadu, India
raja24rv@gmail.com

Naveen Mohan
Dept. of Electronics & Communication Engineering
Jeppiaar Engineering College
Chennai, Tamil Nadu, India
mohannaveen53@gmail.com

Anandhanarayanan Kamalakannan
Central Electronics Engineering Research Institute (Chennai Centre)
CSIR Madras Complex
Taramani, Chennai, Tamil Nadu, India
ananthanalwar@gmail.com

Satish Bindal
Central Electronics Engineering Research Institute (Chennai Centre)
CSIR Madras Complex
Taramani, Chennai, Tamil Nadu, India
satish.bindal@gmail.com

Anita Titus
Dept. of Electronics & Communication Engineering
Jeppiaar Engineering College
Chennai, Tamil Nadu, India
anitatitus72@gmail.com

***Abstract*— Manual inspection of fruit surface defects is time consuming, involves labour cost, prone to human error and possess inconsistent classification standards in fruit sorting application. To solve this issue, an automatic fruit sorting algorithm using deep learning technique was proposed to identify surface defects namely splitting, pitting, and stem-end rot found in mandarin fruits. The fruit sorting algorithm consists of K-Medoids segmentation and Convolutional Neural Network (CNN) classification model. The grayscale images of mandarin fruit surface were captured from an image acquisition system built with Near Infrared (NIR) camera. A preprocessing median filter was applied to remove random noise. After preprocessing, segmentation was carried out using K-Medoids clustering to crop the fruit surface image from the background region. Different CNN models namely VGG-16, InceptionV3 and MobileNet were trained and tested with and without transfer learning approach using the cropped image dataset. After training, the cropped fruit surface image was given to CNN model for defect classification. The classification results of the above models improved significantly after implementing transfer learning method. The VGG-16 model achieved a maximum overall classification accuracy of 90% without transfer learning and 99.53% with transfer learning approach when compared with the InceptionV3 and MobileNet. Overall accuracy of MobileNet improved from 57% to 98% after transfer learning and also it takes minimum time for inference. Considering both the overall accuracy and inference time parameters with the transfer learning approach, the MobileNet is found to be the best model for mandarin fruit sorting application.**

Keywords*—Fruit Sorting, K-Medoids, Convolutional Neural Network, VGG-16, InceptionV3, MobileNet, Transfer learning, Surface defect classification.

I. INTRODUCTION

Agricultural products like fruits, vegetables and food grains are graded into quality categories before being packed or being sent for process line. Traditional grading and sorting involve trained humans, which is time consuming and subjective. To improve the quality and to reduce cost, the food processing industry needs flexible and intelligent inspection systems. Automatic grading and sorting incorporate Machine Vision technology encompassing the image processing techniques by replacing the conventional sensor-based instrumentation. In the past few decades, image processing techniques have been employed for the quality evaluation of the fruits. Machine vision applications make use of the images of the agricultural products being graded which forms non-destructive way of testing [1]. Fruits like lemons, oranges, limes, mandarins and tangerines come under the category of citrus varieties and are grown on a very large scale globally. The citrus fruit defects and diseases may occur due to post harvest handling or due to peel disorders [2]. The major post-harvest diseases found in citrus fruits are stem-end rot, splitting, pitting, green and blue mold, sour and brown rots, anthracnose and alternaria rot, etc. [3]. The external defects and diseases of the citrus fruits affects the quality and appearance and hence influence the consumers acceptancy.

Imaging-based analysis and detection of citrus fruit defects and diseases has been a matter of research study for many years. A variety of methods using machine learning and pattern detection were developed to increase the recognition rate of the different diseases [4]. In recent years, deep learning algorithms were widely used in food inspection applications. In this research work, mandarin fruit image dataset was generated for three types of surface defects. A novel region-based clustering technique called K-Medoids clustering is proposed to segment the fruit images and three distinct convolutional neural network (CNN) models viz. VGG-16, InceptionV3, and MobileNet were utilized to classify three surface defects present on the mandarin fruit images. The classification results and performance evaluation of each CNN model were compared and reported.

This paper discusses about the literature review in Section-II. The proposed mandarin fruit sorting system with a deep learning algorithm is discussed in Section-III. The algorithm implementation details are explained in Section-IV. The results and discussions are briefed in Section-V. The conclusion is given in Section-VI.

II. LITERATURE REVIEW

Machine vision systems are considered to be potential in agriculture, fruit and vegetable industry, etc. Image processing has been proved to be an effective method for analysis in food processing applications. The disease identification of fruits is a critical issue and advanced automatic detection systems need to be developed. With the development of machine learning and deep learning, it is possible to identify and even control crop diseases by using electronic devices instead of manual observation.

Chandan Kumar, et al [5] had developed image processing algorithms to classify the citrus fruits like orange, sweet-lime and lemon using gray level co-occurrence matrix (GLCM) features like contrast, energy, correlation, homogeneity. Giacomo Capizzi, et al [6] presented a machine learning approach to classify fruit surface defects by extracting colour and GLCM based texture features from fruit images. The extracted features from fruit surface area are classified with Radial Basis Probabilistic Neural Network (RBPNN). Blasco et al [7] found that the identification and detection accuracy of citrus fruits with anthracnose increased from 86% to 95% using NIR images when compared with other images.

Mukesh Kumar Tripathi, Dhananjay D. Maktedar [8] explained the role of computer vision in food quality inspection especially in fruits and vegetables. The authors have discussed about a generic computer vision framework to grade the fruits and vegetables with multiple surface defects. Vijaykumar, et al [9] has trained Residual Network CNN model for detecting mellowness found in dragon fruits. Haiyan Zhou, et al [10] proposed a CNN model with W-Softmax as loss function to classify multiple surface defects viz. rot, spot, scar, crack, and normal found in green plums. Nazrul Ismail, et al [11] have focussed to develop a real-time visual inspection system for fruit sorting application using deep learning technique. Yanfei li, et al [12] proposed different methods for apple quality classification containing non-uniform background. Here three methods namely Google InceptionV3 CNN model, HOG+SVM and GLCM+SVM were trained and validated to identify apple quality. Anuja Bhargava and Atul Bansal [13] highlighted the use of image processing based Non-destructive testing for quality inspection of agriculture products. Andrei-Alexandru Tulbure, et al [14] reviewed different defect detection techniques and found that deep learning methods achieved state-of-the-art performance compared to traditional computer vision algorithms.

Most of the previous works are based on Fuzzy C-Means, K-Means, and K-Nearest Neighbour for identification and quality analysis of fruits. The literature survey reveals that the use of automated detection and classification systems for citrus fruit diseases require further development. Hence new tools & enhanced image processing techniques are needed for automatic detection and classification by deploying advanced deep learning methods.

III. PROPOSED METHOD

In this paper, a deep learning-based fruit surface classification algorithm was proposed and implemented using Python, OpenCV, and Keras library. In the proposed algorithm, the clustering technique isolates the fruit surface region from the background and trained convolutional neural network (CNN) model classifies the images precisely. Three distinct CNN models viz. VGG-16, InceptionV3, and MobileNet were chosen to classify no defect and three types of surface defects present on mandarin fruit images. The mandarin fruit surface types viz. no defect, splitting (mechanical damage), pitting, and stem-end-rot are shown in Fig.2(a).

A. Image Acquisition System

An image acquisition system consists of a tabletop conveyor, monochrome NIR camera, ring light source for illuminating fruit samples, and Intel PC system for image processing are shown in Fig.1. The acquisition system is equipped with the IDS camera UI-3240CP which contains a 1.3-megapixel CMOS sensor with a 5.3μm pixel size [15]. The wavelength range of NIR camera is 400-1050nm. The camera has a resolution of 1280 x 1024 pixel at a frame rate of 60 fps. The camera is connected to PC via USB3.0 interface and it is operated using proprietary software. The monochrome image of fruit samples with different surface defects captured with NIR camera are shown in Fig.2(b).

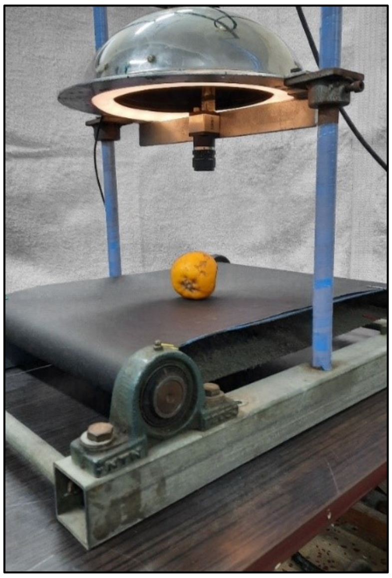

Fig. 1. Image acquisition system with ring light source

B. Image Dataset Preparation

The grayscale image dataset for the proposed work was captured from 200 mandarin fruit samples kept at different orientations using the image acquisition system (Section III A). To eliminate noise, a low pass median filter was applied to the captured image. Then the preprocessed images were resized to 256 × 256 pixels and stored in **.bmp** file format. The fruit surface image dataset consists of 1939 images representing four classes namely no defect, splitting, pitting, and stem-end-rot. The dataset has 378 images of No Defect, 486 images of splitting, 859 images of pitting, and 216 images of stem-end rot. Table I, shows the total number of fruit images that were used for training and testing.

TABLE I. FRUIT IMAGE DATASET

Type of Fruit Surface	Training Samples	Testing samples
No Defect	294	84
Splitting	378	108
Pitting	673	186
Stem-end rot	168	48
Total no. of images	1513	426

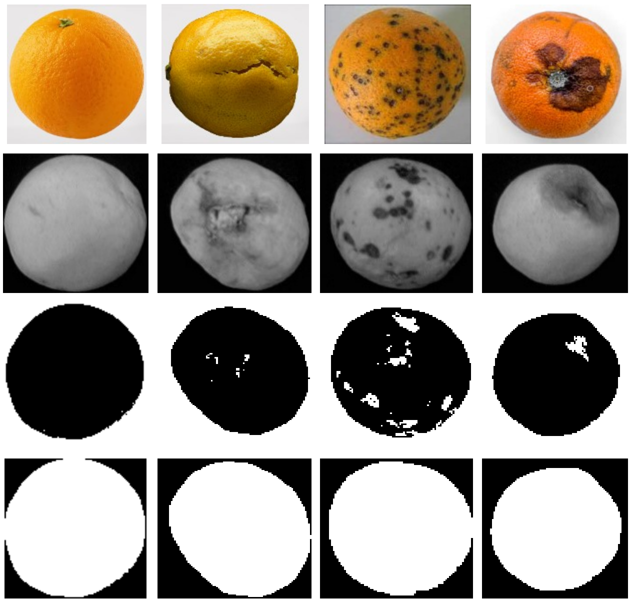

Fig.2. (a) Color image of mandarin fruits showing the surface defect types: no defect, splitting, pitting & stem-end rot (Courtesy: downloaded online) (b) Monochrome image of mandarin fruits captured with NIR enhanced industrial camera. (c) Results of Image segmentation using K-Medoids Clustering. (d) Binary mask image obtained after morphological operation.

C. Image Segmentation using Clustering technique

K-Medoids clustering-based segmentation is applied to the grayscale image for extracting the fruit surface region from the background. When compared with K-Means, K-Medoids is found to be more robust to noise and outliers [16,17]. The K-Medoids algorithm selects one real data point from a given dataset to represent each cluster center, whereas K-Means algorithm takes the average of associated data points in a given dataset to represent the cluster centers. Thereby the K-Medoid technique provides a better interpretation of the cluster centers than K-Means. The block schematic of the segmentation process is shown in Fig.3. The K value represents the number of clusters to be formed when using the K-Medoids clustering technique. After initializing K=2, the grayscale fruit image is given to the K-Medoids clustering algorithm for fruit surface segmentation. Multiple morphological operations viz. erosion and dilation were conducted on the K-Medoids output to get an exact binary mask of the fruit surface. The binary mask is used to extract the fruit surface image from the input image using the bitwise AND operator. As a result, the fruit surface image is cropped from the background region. The output of K-Medoids clustering and binary mask image generated after morphological operations are shown in Fig.2(c) & Fig.2(d) respectively.

D. Deep Learning based fruit sorting algorithm

The image acquisition system generates the fruit image dataset comprising all types of surface defects. As a preprocessing step, a median filter was applied to remove noise present in the captured image. After applying the filter, K-Medoids clustering is used to segment the fruit surface images from the background region. The cropped image dataset obtained from segmentation was given to the CNN model for training. For surface defect classification three different CNN models viz. VGG-16, InceptionV3, and MobileNet were chosen. After training, the trained CNN models were used to classify the fruit surface images as no defect, splitting, pitting and stem-end-rot. The flowchart of fruit sorting algorithm pertaining to mandarin surface defect classification is shown in Fig.4. The classification results of CNN models were analyzed and the best one was chosen as briefed in Section V.

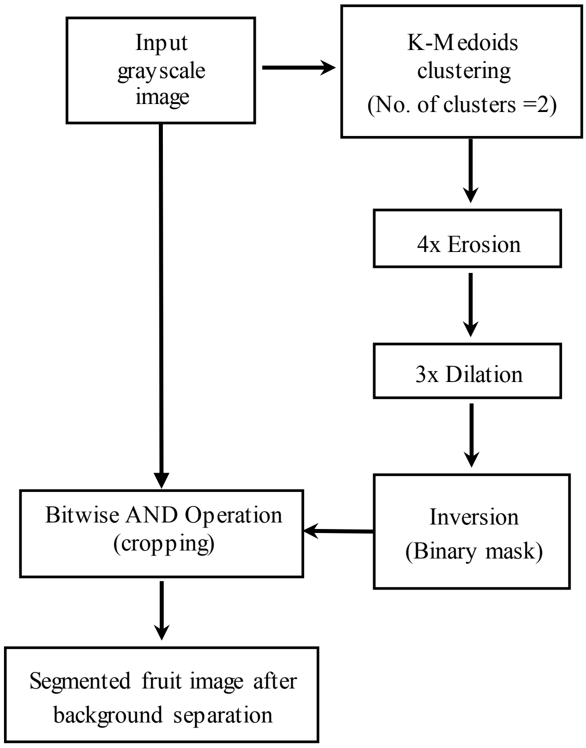

Fig. 3. Image segmentation using K-Medoids Clustering

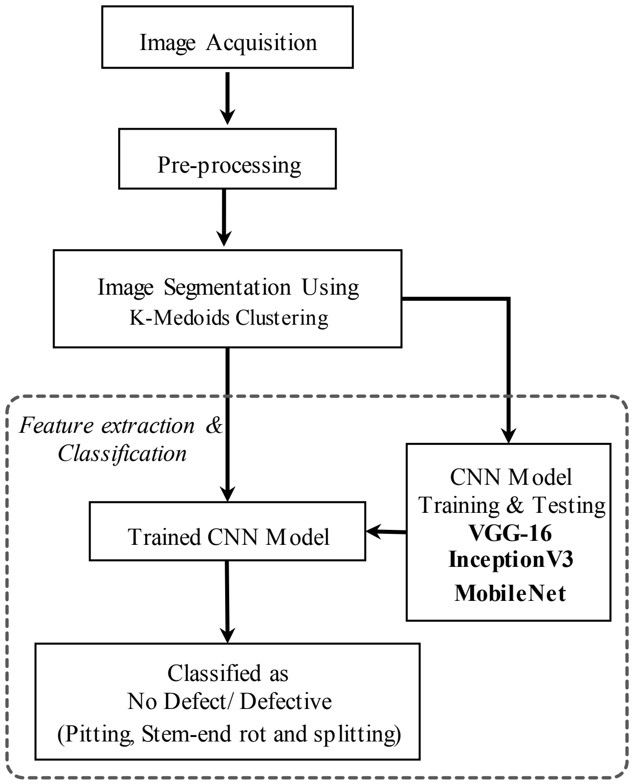

Fig. 4. Block diagram of proposed fruit sorting system

IV. IMPLEMENTATION DETAILS

A total of 1939 fruit image samples in the image dataset were distributed among training and testing phases. A split ratio of approximately 80:20 was chosen to redistribute date among training and testing dataset. The training and testing dataset consist of 1513 and 426 image samples respectively as indicated in Table I. In order to train the CNN models for surface defect classification, the image dataset is categorized according to the type of defect and stored in separate folders. Three CNN models viz. VGG-16, InceptionV3, and MobileNet were chosen and trained using *Adam* optimizer with a learning rate of 0.00001. All the CNN models were trained approximately for 20 epochs with 100 iterations per epoch. For all CNN models, categorical cross entropy was chosen as a loss function as fruit surface sorting is a multi-class classification task. The loss function calculates the cross-entropy between the class probability distributions and its increases when the predicted probability diverges from the desired output. The CNN model training and testing were carried out on an Intel(R) core (TM) 2Quad CPU running at 2.66GHz with 16GB of RAM and NVIDIA Quadro K620 GPU card using the Keras library. A Graphical user interface (GUI) for the fruit sorting system was developed with the Tkinter library as shown in Fig.5. The GUI has provision for CNN model loading, image capturing, and displaying the result along with different image processing outputs.

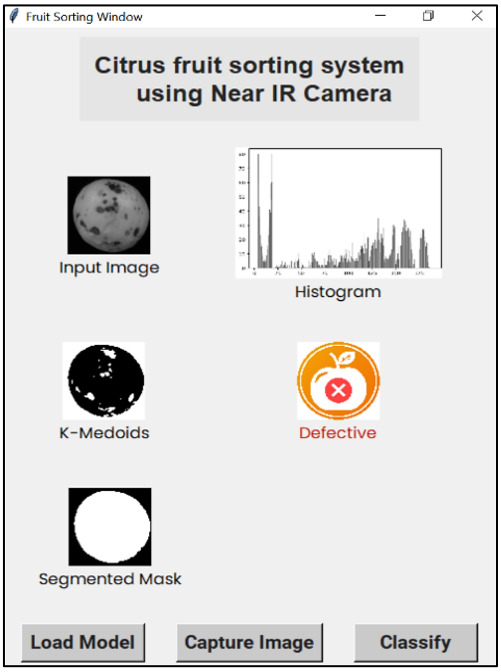

Fig. 5. GUI interface for fruit sorting system

Initially each CNN model was trained by randomly generated weights and the results obtained without transfer learning are tabulated as shown in Table II. To improve the performance of each model, transfer learning technique was applied. In transfer learning, a pre-trained model [18] was chosen as initial weights to further train the CNN model with the training image dataset without changing the model tuning parameters like optimizer, learning rate and loss function. After transfer learning, the CNN algorithms achieved maximum training and testing accuracy for all kinds of surface defects. Finally, all the trained models were saved in hierarchical data format (**. hdf5**).

The testing dataset consists of 426 image samples in which 84,108,186, and 48 belongs to No defect, splitting, pitting, and stem-end rot respectively. During the testing phase, the trained models were used to classify these samples into four classes.

V. RESULTS AND DISCUSSION

The proposed deep-learning based fruit sorting algorithm uses three different CNN model namely VGG-16, InceptionV3 & MobileNet to identify and classify the fruit surface defects. The chosen image dataset consists of 4 classes namely no defect, splitting, pitting & stem-end rot. The performance of three different CNN models with and without transfer learning were compared. The model classification accuracies for different surface defects were varying from 67% to 96% without transfer learning as shown in Table II. After transfer learning, classification accuracies of all defects were found to be more than 98% as shown in Table III.

The transfer learning approach significantly improves the model performance [19] in terms of precision, recall and F1-score for each defect class. Here TP, TN, FP FN and MR represent true positive, true negative, false positive, false negative and misclassification rate per class respectively. The equations for overall accuracy, class accuracy, precision, recall and F1-score are given below.

$$Precision = \frac{TP}{(TP+FP)} \quad (1)$$

$$Recall = \frac{TP}{(TP+FN)} \quad (2)$$

$$Overall\ Accuracy = \frac{TP+TN}{(TP+FP+TN+FN)} \quad (3)$$

$$Class\ Accuracy = 1 - MR \quad (4)$$

$$MR = \frac{FP+FN}{Total\, no.\, of\ all\ class\ samples} \quad (5)$$

$$F1\, Score = 2 * \frac{(Precision * Recall)}{(Precision + Recall)} \quad (6)$$

TABLE II. PERFORMANCE OF DIFFERENT CNN MODELS APPLIED FOR FRUIT SURFACE CLASSIFICATION ***WITHOUT TRANSFER LEARNING***

CNN	Type of defect	Class Accuracy (%)	Precision	Recall	F1 Score
VGG-16	No defect	100	1	1	1
	Splitting	90.61	0.79	0.86	0.82
	Pitting	96.24	0.95	0.97	0.96
	Stem-end Rot	93.43	0.79	0.56	0.66
InceptionV3	No defect	94.84	0.80	0.98	0.88
	Splitting	85.45	0.75	0.64	0.69
	Pitting	91.55	0.93	0.87	0.90
	Stem-end Rot	92.49	0.64	0.77	0.70
MobileNet	No defect	88.50	0.75	0.63	0.68
	Splitting	67.61	0.32	0.25	0.28
	Pitting	70.66	0.62	0.87	0.72
	Stem-end Rot	87.79	0.30	0.063	0.10

TABLE III. PERFORMANCE OF DIFFERENT CNN MODELS APPLIED FOR FRUIT SURFACE CLASSIFICATION ***WITH TRANSFER LEARNING***

CNN	Type of defect	Class Accuracy (%)	Precision	Recall	F1 Score
VGG-16	No defect	100	1	1	1
	Splitting	100	1	1	1
	Pitting	99.53	1	0.99	0.99
	Stem-end Rot	99.53	0.96	1	0.98
InceptionV3	No defect	99.3	0.99	0.98	0.98
	Splitting	99.3	0.99	0.98	0.99
	Pitting	99.53	0.99	1	0.99
	Stem-end Rot	100	1	1	1
MobileNet	No defect	100	1	1	1
	Splitting	100	1	1	1
	Pitting	98.83	0.98	0.99	0.99
	Stem-end Rot	98.83	0.96	0.94	0.95

In both with and without transfer learning methods, VGG-16 model possesses maximum overall classification accuracy of 99.53% and 90.14% respectively as shown in Table IV. Similarly, InceptionV3 model has accuracy of 99.06% & 82.62% respectively. After transfer learning, overall classification accuracy of MobileNet improved from 57.27% to 98.83% as shown in Table IV. It was found that the time taken per inference is minimum in case of MobileNet as referred in Table V. The MobileNet performs well in terms of both overall accuracy and time taken per inference when compared with VGG-16 and InceptionV3. Thus, MobileNet is found to be a feasible model for implementing it for fruit sorting application.

TABLE IV. OVERALL CLASSIFICATION ACCURACY OF DIFFERENT CNN MODELS

CNN Model	Without Transfer Learning (%)	With Transfer Learning (%)
VGG-16	90.14	99.53
InceptionV3	82.62	99.06
MobileNet	57.27	98.83

TABLE V. COMPARISON OF DIFFERENT CNN ARCHITECTURES

Architecture	Input size	Layers	Para-meters (Millions)	Time taken per Inference (ms)
VGG-16	150 x 150 x3	16	138	87
InceptionV3	150 x 150 x3	189	24	142
MobileNet	160 x 160 x3	55	4	54

VI. CONCLUSION

This paper discusses the mandarin fruit sorting system using a deep learning-based fruit surface defect classification algorithm. The proposed fruit sorting algorithm consists of K-Medoids segmentation and a CNN classification model. The K-Medoids clustering technique was chosen to segment the fruit surface portion from the background region. Three CNN models namely VGG-16, InceptionV3, and MobileNet were chosen and trained to classify the fruit surface image into four different categories: No defect, splitting, pitting, and stem-end rot. Initially, all the three CNN models were trained with random weights and their results were tabulated. To improve their performance, again all three models were trained using a transfer learning approach with pre-trained weights. After training, the performance of each CNN classification model was evaluated with and without transfer learning approaches. The results of the classification algorithm significantly improved after implementing transfer learning. The VGG-16 possess maximum overall classification accuracy for both approaches when compared with the other two models. While considering different parameters like overall accuracy, inference time, and memory space, MobileNet was found to be a reliable and efficient model for mandarin fruit sorting application. This method can be extended to other fruit surface sorting applications like apple and similar citrus varieties.

ACKNOWLEDGMENT

The authors are sincerely thankful to the Director, CSIR-CEERI, Pilani and Scientist-in-charge, CEERI Chennai for providing an opportunity to do this work at Chennai Centre.

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